

Question ID d03e29f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: d03e29f1

$$(x - 6)^2 + (y - 3)^2 = 81$$

The graph of the given equation in the xy -plane is a circle. What is the length of the radius of this circle?

- A. 3
- B. 6
- C. 9
- D. 81

ID: d03e29f1 Answer

Correct Answer:

C

Rationale

Choice C is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . The graph of the given equation, $(x - 6)^2 + (y - 3)^2 = 81$, is a circle in the xy -plane. This equation can be written as $(x - 6)^2 + (y - 3)^2 = 9^2$, where $h = 6$, $k = 3$, and $r = 9$. Therefore, the radius of this circle is **9**.

Choice A is incorrect. This is the y -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice B is incorrect. This is the x -coordinate of the center, not the radius, of the circle defined by the given equation.

Choice D is incorrect. This is the value of the radius squared, not the radius, of the circle defined by the given equation.

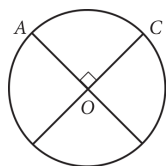
Question Difficulty:

Easy

Question ID 23c5fcce

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ID: 23c5fcce



The circle above with center O has a circumference of 36.
What is the length of minor arc $\widehat{AC}$?

- A. 9
- B. 12
- C. 18
- D. 36

ID: 23c5fcce Answer

Correct Answer:
A

Rationale

Choice A is correct. A circle has 360 degrees of arc. In the circle shown, O is the center of the circle and $\angle AOC$ is a central angle of the circle. From the figure, the two diameters that meet to form $\angle AOC$ are perpendicular, so the measure of $\angle AOC$ is 90° .

Therefore, the length of minor arc $\widehat{AC}$ is $\frac{90}{360}$ of the circumference of the circle. Since the circumference of the circle is 36, the length of minor arc $\widehat{AC}$ is $\frac{90}{360} \times 36 = 9$.

Choices B, C, and D are incorrect. The perpendicular diameters divide the circumference of the circle into four equal arcs; therefore, minor arc $\widehat{AC}$ is $\frac{1}{4}$ of the circumference. However, the lengths in choices B and C are, respectively, $\frac{1}{3}$ and $\frac{1}{2}$ the circumference of the circle, and the length in choice D is the length of the entire circumference. None of these lengths is $\frac{1}{4}$ the circumference.

Question Difficulty:
Easy

Question ID f2495de4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: f2495de4

What is the value of $\cos \frac{565\pi}{6}$?

- A. $\frac{1}{2}$
- B. 1
- C. $\frac{\sqrt{3}}{2}$
- D. $\sqrt{3}$

ID: f2495de4 Answer

Correct Answer:

C

Rationale

Choice C is correct. The cosine of an angle is equal to the cosine of $n(2\pi)$ radians more than the angle, where n is an integer constant. Since $\frac{565\pi}{6}$ is equivalent to $47(2\pi) + \frac{\pi}{6}$, $\cos(\frac{565\pi}{6})$ can be rewritten as $\cos(47(2\pi) + \frac{\pi}{6})$, which is equal to $\cos(\frac{\pi}{6})$. Therefore, the value of $\cos(\frac{565\pi}{6})$ is equal to the value of $\cos(\frac{\pi}{6})$, which is $\frac{\sqrt{3}}{2}$.

Alternate approach: A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. The cosine of a number t is the x-coordinate of the point resulting from traveling a distance of t counterclockwise from the point (1, 0) around a unit circle centered at the origin in the xy-plane. A unit circle has a circumference of 2π . It follows that since $\frac{565\pi}{6}$ is equal to $47(2\pi) + \frac{\pi}{6}$, traveling a distance of $\frac{565\pi}{6}$ counterclockwise around a unit circle means traveling around the circle completely 47 times and then another $\frac{\pi}{6}$ beyond that. That is, traveling $\frac{565\pi}{6}$ results in the same point as traveling $\frac{\pi}{6}$. Traveling $\frac{\pi}{6}$ counterclockwise from the point (1, 0) around a unit circle centered at the origin in the xy-plane results in the point $(\frac{\sqrt{3}}{2}, \frac{1}{2})$. Thus, the value of $\cos \frac{565\pi}{6}$ is the x-coordinate of the point $(\frac{\sqrt{3}}{2}, \frac{1}{2})$, which is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect. This is the value of $\sin \frac{565\pi}{6}$, not $\cos \frac{565\pi}{6}$.

Choice B is incorrect. This is the value of the cosine of a multiple of 2π , not $\frac{565\pi}{6}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan \frac{565\pi}{6}}$, not $\cos \frac{565\pi}{6}$.

Question Difficulty:

Easy